

CU 323

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operation instructions describe Grundfos CU 323.

Sections 1-2 give the information necessary to be able to install the product in a safe way.

Sections 3-9 give important information about the product, as well as information on service, fault finding and disposal of the product.

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Read this document before installation. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information

1.1 Target group

These installation and operating instructions are intended for professional installers.

1.2 Symbols used in this document

1.2.1 Warnings against hazards involving risk of death or personal injury

DANGER



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

WARNING



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

CAUTION



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

The text accompanying the three hazard symbols DANGER, WARNING and CAUTION is structured in the following way:

SIGNAL WORD

Description of hazard



Consequence of ignoring the warning.
- Action to avoid the hazard.

1.2.2 Other important notes



A blue or grey circle with a white graphical symbol indicates that an action must be taken.



A red or grey circle with a diagonal bar, possibly with a black graphical symbol, indicates that an action must not be taken or must be stopped.



If these instructions are not observed, it may result in malfunction or damage to the equipment.



Tips and advice that make the work easier.

2. Installing the product

CU 323 is only intended for factory wiring.

Before installing the product, check the following:

- CU 323 corresponds to the order.
- CU 323 is suitable for the supply voltage and frequency at the installation site.
- CU 323 has not been damaged during transportation.

DANGER

Electric shock



Death or serious personal injury

- Switch off the power supply before you start any work on the product. Make sure that the power supply cannot be accidentally switched on.

The installation must be carried out by authorised staff in accordance with local regulations.

Observe all safety regulations at the installation site.

Be aware that terminals L and N as well as 70 to 81 may be connected to dangerous contact voltage. External control voltage from other groups may occur.

DANGER

Electric shock



Death or serious personal injury

- The installation must incorporate a main switch. The main switch must be placed close to CU 323 and easily accessible for the operator. The switch must be marked as main switch for CU 323. The main switch must be according to IEC 60947-1 and IEC 60947-3.

WARNING

Electric shock



Death or serious personal injury

- All wires to units outside the control panel must be of the type H05VV-F according to CENELEC HD21. USA and Canada: All wires must comply with the National Electrical Code (NEC) and/or Canadian Electrical Code.

2.1 Location



Install CU 323 in a well-ventilated location, and do not place it close to any heat sources.

CU 323 is designed for indoor installation in the front of a control panel or a separate cabinet.

For outdoor installation, mount CU 323 in a control cabinet with an enclosure class of minimum IPX4.

2.1.1 Potentially explosive environments

CU 323 must not be installed in potentially explosive environments, but may be used together with Grundfos pumps approved for installation in potentially explosive environments.

2.2 Mechanical installation

Fasten CU 323 with the four M5 x 10 screws supplied with the product. See fig. 1 (1).

Maximum torque: 1.4 Nm.

For dimensions of CU 323, see section 9. *Disposing of the product.*

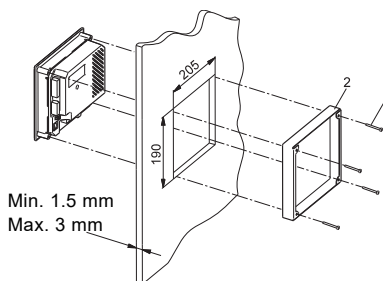


Fig. 1 Mounting in panel front, standard

Pos.	Description
1	Screw (4 pcs)
2	Mounting frame

Connect the earth terminal between the cable clamps on CU 323 to the mounting frame. See fig. 1 (2) and fig. 2.

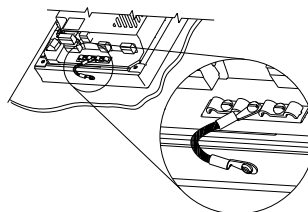


Fig. 2 Earth connection

TM03 1301 4105

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2.3 Electrical installation

2.3.1 EMC-correct installation

CU 323 is normally mounted in a panel which also contains contactors and other power equipment.

To ensure faultless function, install the electronic modules in an EMC-correct way:

- Ensure sufficient earth connection between CU 323 and the frame. See fig. 2.
- Use screened cables for internal GENibus. Connect the screen to the cable clamp in front of the terminals A1, Y1 and B1. See fig. 3.
- Use screened cables for CIM modules. Connect the screen to the cable clamp in front of the CIM module terminals. See fig. 3.

Terminals for CIM module

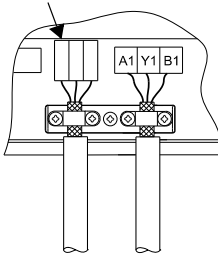


Fig. 3 Cable screens for external and internal GENibus connections fixed with cable clamps

- Use screened signal conductors for digital and analog inputs.



Remove any isolating plastic tape between the screen and the sheath before mounting the cable in the cable clamp.



Do not twist the screen ends as this will destroy the screening effect at high frequencies.

2.3.2 GENibus connection to E-pumps

CU 323 can communicate with one or more E-pumps via internal GENibus.

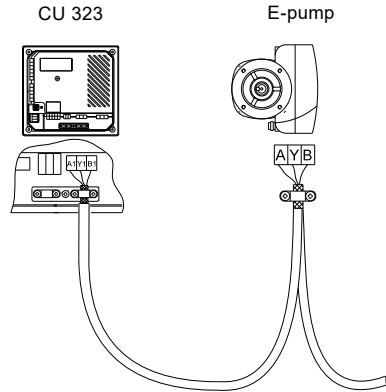


Fig. 4 GENibus connection to E-pumps

Connection with variable frequency drive via Modbus (RS485):

GENibus terminal	Description
A	Positive
Y	Common
B	Negative

2.3.3 Fieldbus communication interface modules (CIM)

CU 323 can be connected to an external communication network via an add-on fieldbus CIM module.

A CIM module is a communication interface module.

These CIM modules are available:

- CIM 050 RS-485 GENI
- CIM 110 LON
- CIM 150 Profibus
- CIM 200 Modbus RTU
- CIM 250 GSM/GPRS
- CIM 300 BACnet MS/TP
- CIM 500 Modbus TCP, Profinet or BACnet IP.

The CIM modules must be ordered separately.

The installation of the CIM module is described in the Hydro Multi-B service instructions available in Grundfos Product Center. Copy the link below to see the document online:

<http://net.grundfos.com/qr/i/97916525>.

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TM04 9276 3810

3. Product introduction

3.1 Product description

The CU 323 control unit is designed for the monitoring and control of up to four pumps.

3.2 User interface

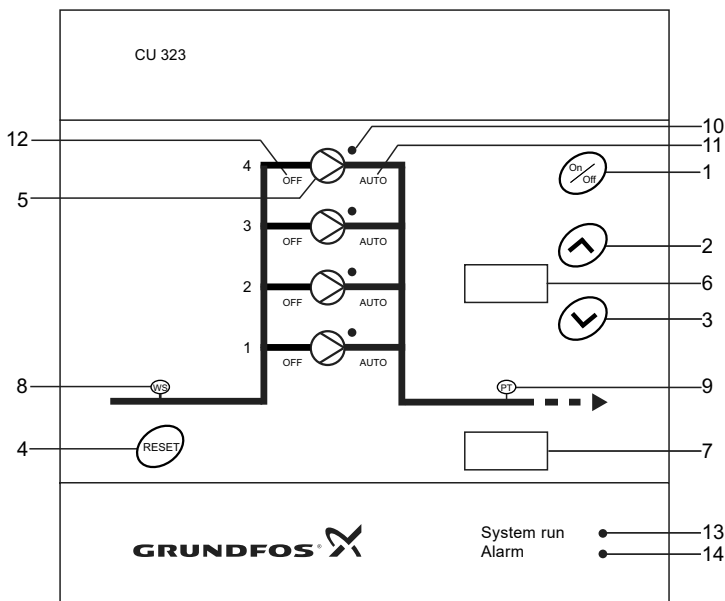


Fig. 5 CU 323 user interface

Pos.	Buttons	Description
1		[On/Off]: Changes the system operating mode between "Normal" and "Stop".
2		[Up]: Increases the setpoint value.
3		[Down]: Decreases the setpoint value.
4		[RESET]: Resets all alarms.
5		[>]: Changes the operating mode of the individual pumps between "AUTO" and "OFF". Each pump in the system has an Auto/Off button.



For further information about how to operate CU 323, see section [3.3 Layout of the back of CU 323](#).

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Pos.	Display	Description
6	"Setpoint"	Shows the setpoint. After unlocking CU 323, the setpoint will flash.
	<i>PPP</i>	Proportional pressure control. "PPP" will be flashing for one second followed by the current setpoint for four seconds. This function is only available for E-pump systems.
	<i>l _ _ l</i>	The tank symbol appears in tank-filling mode with float switches.
	<i>l _ _ l</i> or "Setpoint"	Alternates between the tank symbol and the setpoint in tank-filling mode with analog input.
	<i>E.St</i>	External stop: Indicates that the system has been stopped externally via a digital input.
	<i>OFF</i>	Indicates that the system has been set to "OFF".
	<i>On</i>	Indicates that the system has been set from "OFF" to "On". "On" is only shown for three seconds after which the setpoint is shown.
	<i>0-7</i>	Indicates that the buttons on CU 323 have been locked automatically. To unlock the buttons, press [RESET] and arrow up or [RESET] and arrow down simultaneously.
	<i>bUS</i>	Indicates that the system is remotely controlled.
	<i>LOC</i>	Local: Appears if attempts to change the local settings are made while the system is remotely controlled.
7	"Measured value"	Shows the actual, measured value on the outlet side.
	<i>PPS</i>	Proportional setpoint at zero flow, PPS [bar]. "PPS" is visible if you press [RESET] for four seconds. After [RESET] has been released, you can adjust "PPS" by means of the arrow up or arrow down buttons. If the buttons are not activated for four seconds, the system returns to normal operation. This function is only available for E-pump systems.
	<i>- - -</i>	Appears in tank-filling mode with float switches.
	<i>Err</i>	Appears in case of a primary sensor fault.
	<i>H.L</i> or "Measured value"	Alternates between the actual, measured value on the outlet side and "H.L" when a maximum outlet pressure or a level limit has been exceeded.

Pos.	Indicator light	Status	Description
8		Off	Normal state.
		Red	Water shortage.
9		Off	Normal state.
		Red	Fault, primary sensor and/or redundant sensor. The maximum outlet pressure or the level limit has been exceeded.
10	 Pump LED	Off	The pump has stopped due to cascade operation. The system is in "OFF" state. The power supply has been switched off.
		Red	A pump fault has occurred.
		Green	The pump is running.
		Flashing green	The pump is in standby mode or has been stopped by pressing [>] or via remote control.
11	AUTO	Yellow	The pump is controlled by CU 323.
12	OFF	Yellow	The pump has been stopped by pressing [>] or via remote control.

Pos.	Indicator light	Status		Description
		Pos. 13: "System run" (green)	Pos. 14: "Alarm" (red)	
13 and 14	 System LEDs	Off	Off	The power supply has been switched off.
		On	Off	The system is in normal operation.*
		Flashing	Off	The system has been set to stop.
		Off	On	The system has been stopped due to an alarm.
		On	On	The system is in normal operation, but there is a warning.*
		Flashing	On	The system has been set to stop, and there is a warning.

* The last pump in operation might have been stopped temporarily by the low-flow stop function.

3.3 Layout of the back of CU 323

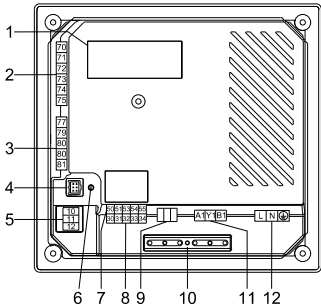


Fig. 6 Back of CU 323

Pos.	Description
1	Nameplate
2	Terminals for relay outputs
3	Terminals for digital outputs
4	Service connection
5	Terminals for digital inputs
6	Voltage indicator
7	Terminals for analog inputs
8	Terminals for PTC inputs
9	Terminals for CIM module, not standard
10	Cable clamps for GENIbus connection
11	Internal GENIbus connection
12	Terminals for power supply

3.4 Identification

3.4.1 Nameplate

CU 323 can be identified by means of the nameplate on the back. See fig. 7.

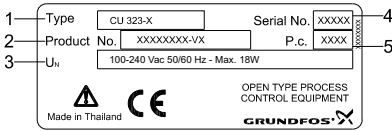


Fig. 7 Nameplate

Pos.	Description
1	Type designation
2	Product number and version number
3	Rated voltage, frequency and power
4	Serial number
5	Production code (year, week)

3.4.2 Type key

Code	Example	CU	32	3	-	4
CU	Control unit					
32	Controller series					
3	Model number					
4	Number of pumps (2 to 4)					

3.4.3 Protective earth terminal

The protective earth terminal is marked with the symbol below. See fig. 8.



Fig. 8 Protective earth terminal

4. Control functions

4.1 Functions of indicator lights and relays

System indicator lights		Operating relay	Alarm relay	Description
"System run" (green)	"Alarm" (red)			
Off	Off			The power supply has been switched off.
Permanently on	Off			The system is in normal operation, and one or several pumps are running.
Permanently on	Off			The system is in normal operation, but the last pump in operation has been stopped by the low-flow stop function.
Flashing	Off			The system has been set to stop.
Off	Permanently on			The system has been stopped due to an alarm.
Permanently on	Permanently on			The system is in normal operation, but there is a warning.
Flashing	Permanently on			The system has been set to stop, and there is a warning.

5. Operating the product

User operation depends on the system mode. The system modes are described in the following sections.



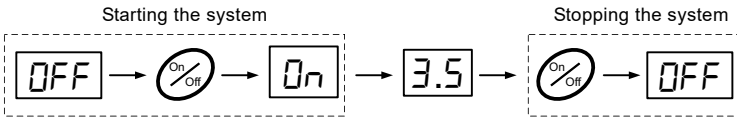
Read this document before startup.
Installation and operation must comply with local regulations and accepted codes of good practice.

5.1 Local mode, external stop not activated

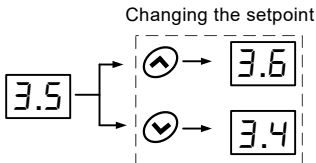
In this mode, the system is locally controlled, and the external stop via digital input is not activated.

The flow charts below show examples of how settings can be made.

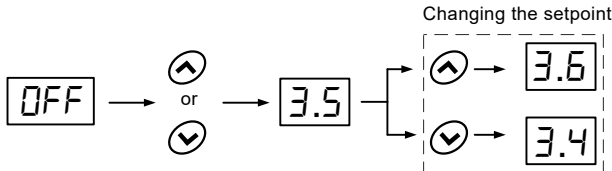
Starting and stopping the system



Changing the setpoint while the system is running



Changing the setpoint while the system is in "OFF" mode



5.2 Local mode, external stop activated

In this mode, the system is locally controlled, and the external stop via digital input is activated.

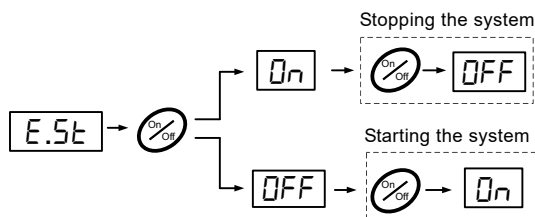
External stop via digital input has higher priority than local control. This means that settings can be changed locally, but they will not be effective until external stop has been deactivated. The flow charts below show examples of how settings can be made.

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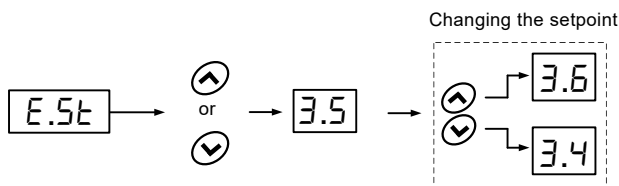
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Stopping and starting the system



If you do not touch the buttons for three seconds, CU 323 reverts to the initial display.

Changing the setpoint

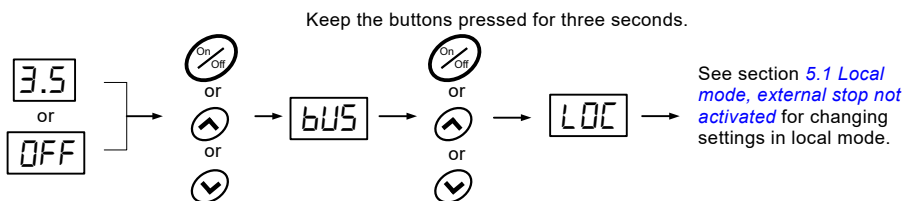


If you do not touch the buttons for three seconds, CU 323 reverts to the initial display.

5.3 Remote-controlled, external stop not activated

In this mode, the system is remote-controlled, and the external stop via digital input is not activated.

If you want to change the local settings, follow the flow chart examples below.



If you do not touch the buttons for three seconds, CU 323 reverts to the initial display.

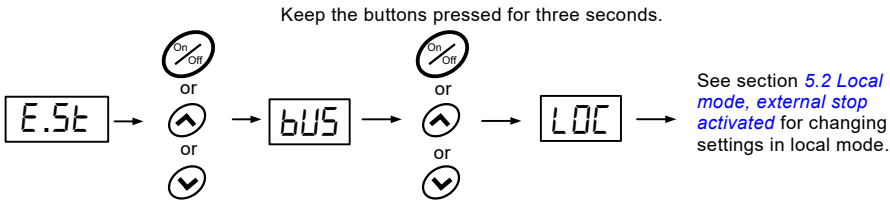
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TM04 9582 4610

5.4 Remote-controlled, external stop activated

In this mode, the system is remote-controlled, and the external stop via digital input is activated.
If you want to change the local settings, follow the flow chart examples below.



If you do not touch the buttons for three seconds, CU 323 reverts to the initial display.

5.5 Adjusting the proportional pressure setpoint

The function can only be enabled in pressure-controlled systems and automatically adapts the setpoint to the actual flow rate to compensate for flow-dependent dynamic losses.
The adaptation can only be square. See fig. 9.

A filter factor can be set to prevent fluctuation.

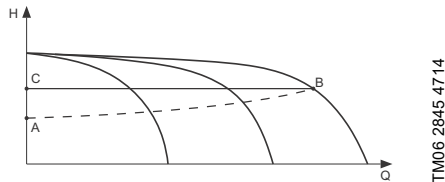


Fig. 9 Proportional pressure

Pos.	Description
A	Pressure at zero flow. Starting point of proportional-pressure control. Influence at 0 flow is equal to x % of the setpoint.
B	Qnom
C	Setpoint

The function has these purposes:

- to compensate for pressure losses
- to reduce the energy consumption
- to increase the comfort for the user.

Activating the function via PC Tool E-products

Settings > Primary controller.

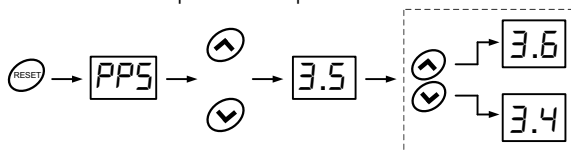
1. Select: Enabled proportional pressure.
2. Set: Influence at zero flow.
3. Set: Filter factor.
4. Select: Hmax / Enter value. Based on pump data.
5. Select: Qnom / Enter value. Based on pump data.
6. Select: Hnom / Enter value. Based on pump data.

Activate the function via the control panel.

This is only applicable when the function is enabled in PC Tool E-products.

1. Press [RESET] for four seconds. Let go of the button when the text "PPS" appears in the display.
2. Adjust the "PPS" value with the arrow up and arrow down buttons.

Keep the buttons pressed for four seconds.



The "PPS" value is defined in bar at zero flow.

6. Servicing the product

6.1 Maintaining the product

CU 323 is maintenance-free during normal use and operation. Clean CU 323 with a wet cloth.

6.2 Repairing the product

CU 323 cannot be repaired. If CU 323 is faulty, replace the unit.

See section [6.3 Replacing the product](#).

6.3 Replacing the product

1. Switch off the power supply to CU 323.
2. Switch off the power supply to components with external supply.
3. Remove the terminal blocks from the sockets.
4. Remove CU 323 from the panel or cabinet.
5. Fit the new CU 323 as described in section [2.2 Mechanical installation](#).
6. Configure the new CU 323 by means of PC Tool E-products.

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7. Fault finding the product

DANGER

Electric shock

Death or serious personal injury



- Switch off the power supply and wait at least five minutes before you make any connections in the pumps, terminal boxes or control cabinet. Make sure that the power supply cannot be accidentally switched on.

Fault	Possible cause	Remedy
1. The pumps are not running.	a) The actual pressure is higher than or equal to the setpoint.	Wait until the pressure has dropped, or lower the pressure on the outlet side of the booster system. Check that the pumps start.
	b) The power supply has been switched off.	Connect the power supply.
	c) The main switch has cut out.	Cut in the main switch.
	d) The main switch is defective.	Replace the main switch.
	e) The motor protection has been activated.	Contact Grundfos.
	f) The motor is defective.	Repair or replace the motor.
	g) The pressure transmitter is defective.	Replace the pressure transmitter. 0-10 bar or 0-6 bar transmitters with 0-10 V, 0-20 mA or 4-20 mA output signals are monitored by the booster system.
	h) The cable is broken or short-circuited.	Repair or replace the cable.
2. The pumps start, but stop immediately. The operating pressure is not reached.	a) Water shortage or no inlet pressure.	Re-establish the supply of water to the booster system. When the inlet pressure has been re-established, press [RESET] to restart the system.
3. The booster system is stopped and cannot restart.	b) The pressure transmitter is defective.	Replace the pressure transmitter. 0-10 bar or 0-6 bar transmitters with 0-10 V, 0-20 mA or 4-20 mA output signals are monitored by the system.
	c) The cable is broken or short-circuited.	Repair or replace the cable.
	d) The power supply to CU 323 has been switched off.	Reconnect the power supply.
	e) CU 323 is defective.	Contact Grundfos.
4. Unstable water supply from the booster system.	a) The inlet pressure is too low.	Check the inlet pipe and the inlet strainer, if any.
	b) The inlet pipe, strainer or pumps are partly blocked by impurities.	Clean the inlet pipe, strainer or pumps.
	c) The pumps suck air.	Check the inlet pipe for leakages.
	d) The pressure transmitter is defective.	Replace the transmitter.

Fault	Possible cause	Remedy
5. The pumps are running, but deliver no water.	a) The valves are closed.	Open the valves.
	b) The inlet pipe or pumps are blocked by impurities.	Clean the inlet pipe or pumps.
	c) The non-return valve is blocked in the closed position.	Clean the non-return valve. Check that the non-return valve moves freely.
	d) The inlet pipe is leaky.	Check the inlet pipe for leakages.
	e) There is air in the inlet pipe or pumps.	Vent and prime the pumps. Check the inlet pipe for leakages.
6. The booster system is unable to reach the setpoint.	a) The consumption is too high.	• Reduce the consumption, if possible. • Install a bigger booster system.
	b) Too many standby pumps have been selected.	Reduce the number of standby pumps.
	c) There is a pipe fracture or a leakage in the system.	Check the system, and repair the damaged parts, if necessary.
7. Leakage from the shaft seal.	a) The shaft seal is defective.	Replace the shaft seal.
8. Noise.	a) The pumps are cavitating.	Clean the inlet pipe or pumps and possibly the inlet strainer.
9. Very frequent starts and stops.	a) The diaphragm tank precharge pressure is not correct.	Set the correct precharge pressure.

8. Technical data

Transient voltages typically present on the mains supply are category 2.

8.1 Operating conditions

Altitude above sea level

Maximum 2000 m.

Ambient temperature

During operation, in stock or during transportation:
-20 to +60 °C.

Do not expose the product to direct sunlight.

Relative humidity

Maximum 95 %.

Pollution degree

Internal pollution degree 2.

External pollution degree 3.

8.2 Mechanical data

Enclosure class

To reduce the external pollution level to maximum 2, install CU 323 in a protecting environment with minimum IPX4 enclosure according to IEC 60529. The cabinet must be of a flame-retardant material. The front of CU 323 is IP54.



To achieve enclosure class IP54, CU 323 must be mounted in the front of an IP54 enclosure so that the back is protected inside the enclosure. See fig. 10.

USA and Canada

CU 323 is type 3R when mounted in the front of a cabinet with type rating 1, 2, 3, 3R, 5, 12, 12K or 13.

8.3 Electrical data

8.3.1 Supply voltage

1 x 100-240 VAC ± 10 %, 50/60 Hz, PE.

Class 1 equipment.

8.3.2 Circuit breaker

Installation of circuit breakers must comply with local and national regulations.

8.3.3 Power consumption

Maximum 12 W without CIM module fitted.

Maximum 18 W with CIM module fitted.

8.3.4 Digital inputs

Open-circuit voltage	24 VDC
Closed-circuit current	5 mA, DC
Frequency range	0-4 Hz

8.3.5 Analog inputs

Input current and voltage	0-20 mA 4-20 mA 0-10 V
Tolerance	± 3.3 % of full scale
Repetitive accuracy	± 1 % of full scale
Input resistance, current	Below 250 Ω
Input resistance, voltage	10 kΩ ± 10 %
Supply to sensor	24 V, maximum 30 mA per sensor, short-circuit-protected

8.3.6 Digital outputs (relay outputs)

Normally open contacts, DO1, DO2 and DO3	C, NO
Normally closed contacts, RO1 and RO2	C, NC, NO
Maximum contact load	240 VAC, 2 A
Minimum contact load	5 VDC, 10 mA

8.3.7 Inputs for PTC sensor or thermal switch

For PTC sensors to DIN 44082. Thermal switches can also be connected.

Open-circuit voltage	8 VDC ± 15 %
Closed-circuit current	1.7 mA, DC

8.3.8 Conductors

Rigid conductors	0.5 - 2.5 mm ²
Flexible conductors without ferrule	20-13 AWG (US)
Flexible conductors with ferrule	0.5 - 1.5 mm ²
With or without plastic collar	20-13 AWG (US)

8.3.9 Terminals

All terminals are suitable for conductors of 0.5 to 2.5 mm² or AWG 20-13.

For further information, see section

[8.3.8 Conductors.](#)

8.3.10 Terminal groups

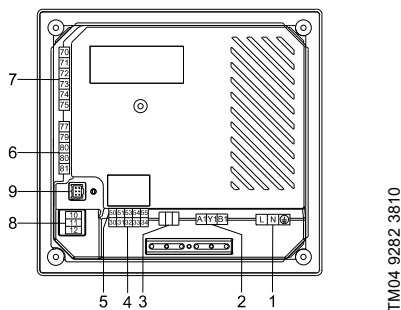


Fig. 10 Terminal groups

The terminals of the groups 4, 6 and 7 are insulated from all other terminal groups by reinforced insulation, 2224 VAC.

Group	Description
1	Connection of power supply
2	Internal GENIbus connection
3	Fieldbus connection for CIM module (optional)
4	Input for PTC sensor or thermal switch
5	Analog inputs
6	Digital outputs
7	Relay outputs
8	Digital inputs
9	Service connection

8.4 Overview of inputs and outputs

Legend

Abbreviation	Description
AI	Analog input
C	Common
DI	Digital input
DO	Digital output
NC	Normally closed contact
NO	Normally open contact
PTC	Positive temperature coefficient
RO	Relay output

For position numbers, see fig. 10.

Pos.	Terminal	Designation	Data	Diagram
1	L	Connection to phase conductor	1 x 100-240 VAC ± 10 %, 50/60 Hz	
	N	Connection to neutral conductor		
	PE	Connection to protective earth		
2	A1	RS-485 A	GENIbus (internal) Fix the screen with a cable clamp.	
	Y1	RS-485 GND		
	B1	RS-485 B		
	⏏	Functional earth		
3	Connection to external fieldbus, see installation and operating instructions for the CIM module.			
4	30	PTC 1	Input for PTC sensor or thermal switch.	
	31	GND, PTC		
	32	PTC 2		
	33	GND, PTC		
	34	PTC 3		
5	50	+24 V	Supply to sensor. Short-circuit-protected 30 mA.	
	51	AI1	Input for analog signal, 0-20 mA, 4-20 mA or 0-10 V.	
	53	+24 V	Supply to sensor. Short-circuit-protected 30 mA.	
	54	AI2	Input for analog signal, 0-20 mA, 4-20 mA or 0-10 V.	
	55	GND		
All terminals, except mains terminals, must only be connected to voltages not exceeding 16 Vrms and 22.6 Vpeak or 35 VDC.				

Pos.	Terminal	Designation	Data	Diagram
6	77	DO1 NO	Relay contact for starting and stopping mains-operated pumps.	
	79	DO2 NO		
	80	C		
	80	C		
	81	DO3 NO		
7	70	Relay 1	C	
	71	Relay 1	NO	
	72	Relay 1	NC	
	73	Relay 2	C	
	74	Relay 2	NO	
	75	Relay 2	NC	
8	10	DI1	Digital input	
	11	GND		
	12	DI2		
All terminals, except mains terminals, must only be connected to voltages not exceeding 16 Vrms and 22.6 Vpeak or 35 VDC.				
9	GENIbus		Service connection	

8.5 Dimensions

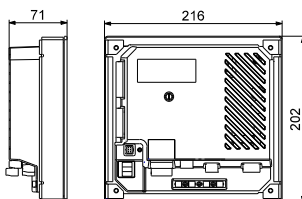


Fig. 11 Dimensional sketch

9. Disposing of the product

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.



The crossed-out wheellie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

TM04 9283 3810

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products CU323, CU352, CU354, CU362, CU372, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

CZ: Prohlášení o shodě EU

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky CU323, CU352, CU354, CU362, CU372, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství.

DK: EU-overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produkterne CU323, CU352, CU354, CU362, CU372 som erklæringen nedenfor omhandler, er i overensstemmelse med Rådets direktiver der er nævnt nedenfor, om indbyrdes tilnærmelse til EU-medlemsstaternes lovgivning.

ES: Declaración de conformidad de la UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos CU323, CU352, CU354, CU362, CU372 a los que hace referencia la siguiente declaración cumplen lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la UE.

FR: Déclaration de conformité UE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits CU323, CU352, CU354, CU362, CU372, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des États membres UE relatives aux normes énoncées ci-dessous.

HR: EU deklaracija sukladnosti

Mi, Grundfos, izjavljujemo s punom odgovornošću da su proizvodi CU323, CU352, CU354, CU362, CU372, na koja se izjava odnosi u nastavku, u skladu s direktivama Vijeća dolje navedene o usklađivanju zakona država članica EU-a.

IT: Dichiarazione di conformità UE

Grundfos dichiara sotto la sua esclusiva responsabilità che i prodotti CU323, CU352, CU354, CU362, CU372, ai quali si riferisce questa dichiarazione, sono conformi alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri UE.

LV: ES atbilstības deklarācija

Sabiedrība Grundfos ar pilnu atbildību paziņo, ka produkti CU323, CU352, CU354, CU362, CU372, uz kuru attiecas tālāk redzamā deklarācija, atbilst tālāk norādītajām Padomes direktīvām par EK/ES dalībvalstu normatīvo aktu tuvināšanu.

PL: Deklaracja zgodności UE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze produkty CU323, CU352, CU354, CU362, CU372, których deklaracja niniejsza dotyczy, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich.

RO: Declarația de conformitate UE

Noi Grundfos declarăm pe propria răspundere că produsele CU323, CU352, CU354, CU362, CU372, la care se referă această declarație, sunt în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre UE.

RU: Декларация о соответствии нормам ЕС

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия CU323, CU352, CU354, CU362, CU372, к которым относится нижеприведенная декларация, соответствующим нижеприведенным Директивам Совета Евросоюза о тождественности законов стран-членов ЕС.

SI: Izjava o skladnosti EU

V Grundfosu s polno odgovornostjo izjavljamo, da je izdelek CU323, CU352, CU354, CU362, CU372, na katerega se spodnja izjava nanaša, v skladu s spodnjimi direktivami Sveta o približevanju zakonodaje za izenačevanje pravnih predpisov držav članic EU.

BG: Декларация за съответствие на ЕО

Ние, фирма Grundfos, заявяваме с пълна отговорност, че продуктите CU323, CU352, CU354, CU362, CU372, за които се отнася настоящата декларация, отговарят на следните директиви на Съвета за уеднавяване на правните разпоредби на държавите-членки на ЕО.

DE: EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte CU323, CU352, CU354, CU362, CU372, auf die sich diese Erklärung beziehen, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen.

EE: EÜ vastavusdeklaratsioon

Meie, Grundfos, kinnitame ja kanname ainuiskulist vastutust selle eest, et toode CU323, CU352, CU354, CU362, CU372, mille kohta all olev deklaratsioon käib, on kooskõlas Nõukogu Direktiividega, mis on nimetatud all pool vastavalt vastuvõetud õigusaktidele ühtlustamise kohta EÜ liikmesriikides.

FI: EU-vaatimustenmukaisuusvakuutus

Grundfos vakuuttaa omalla vastuullaan, että tuotteet CU323, CU352, CU354, CU362, CU372, joita tämä vakuutus koskee, ovat EU:n jäsenvaltioiden lainsäädännön lähentämiseen tähtäävien Euroopan neuvoston direktiivien vaatimusten mukaisia seuraavasti.

GR: Δήλωση συμμόρφωσης ΕΕ

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα CU323, CU352, CU354, CU362, CU372, στα οποία αναφέρεται η παρακάτω δήλωση, συμμορφώνονται με τις παρακάτω Οδηγίες του Συμβουλίου περί προσέγγισης των νομοθεσιών των κρατών μελών της ΕΕ.

HU: EU megfelelőségi nyilatkozat

Mi, a Grundfos vállalat, teljes felelősséggel kijelentjük, hogy a(z) CU323, CU352, CU354, CU362, CU372 termékek, amelyre az alábbi nyilatkozat vonatkozik, megfelelnek az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak.

LT: ES atitikties deklaracija

Mes, Grundfos, su visa atsakomybe pareiškiamo, kad produktai CU323, CU352, CU354, CU362, CU372, kuriems skirta ši deklaracija, atitinka Žemiau nurodytas Tarybos Direktyvas dėl ES šalių narių įstatymų suderinimo.

NL: EU-conformiteitsverklaring

Wij, Grundfos, verklaren geheel onder eigen verantwoordelijkheid dat de producten CU323, CU352, CU354, CU362, CU372, waarop de onderstaande verklaring betrekking heeft, in overeenstemming zijn met de onderstaande Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgeving van de EU-lidstaten.

PT: Declaração de conformidade UE

A Grundfos declara sob sua única responsabilidade que os produtos CU323, CU352, CU354, CU362, CU372, aos quais diz respeito a declaração abaixo, estão em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da UE.

RS: Deklaracija o usklađenosti EU

Mi, kompanija Grundfos, izjavljujemo pod punom vlastitom odgovornošću da je proizvod CU323, CU352, CU354, CU362, CU372, na koji se odnosi deklaracija ispod, u skladu sa dole prikazanim direktivama Saveta za usklađivanje zakona država članica EU.

SE: EU-försäkran om överensstämmelse

Vi, Grundfos, försäkrar under ansvar att produkterna CU323, CU352, CU354, CU362, CU372, som omfattas av nedanstående försäkran, är i överensstämmelse med de rättsdirektiv om inbördes närmande till EU-medlemsstaternas lagstiftning som listas nedan.

SK: ES vyhlásenie o zhode

My, spoločnosť Grundfos, vyhlasujeme na svoju plnú zodpovednosť, že produkty CU323, CU352, CU354, CU362, CU372 na ktoré sa vyhlásenie uvedené nižšie vzťahuje, sú v súlade s ustanoveniami nižšie uvedených smerníc Rady pre zblíženie právnych predpisov členských štátov EÚ.

TR: AB uygunluk bildirgesi

Grundfos olarak, aşağıdaki bildirim konusu olan CU323, CU352, CU354, CU362, CU372 ürünlerinin, AB Üye ülkelerinin direktiflerinin yakınlaştırılmasıyla ilgili durumun aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

CN: 欧盟符合性声明

我们，格兰富，在我们的全权责任下声明，产品 CU323, CU352, CU354, CU362, CU372，即该合格证所指之产品，欧盟使其成员国法律趋于一致的以下理事会指令。

KO: EU 적합성 선언

Grundfos는 아래의 선언과 관련된 CU323, CU352, CU354, CU362, CU372 제품이 EU 회원국 법률에 기반하여 아래의 이사회 지침을 준수함을 단독 책임 하에 선언합니다.

ID: Deklarasi kesesuaian Uni Eropa

Kami, Grundfos, menyatakan dengan tanggung jawab kami sendiri bahwa produk CU323, CU352, CU354, CU362, CU372, yang berkaitan dengan pernyataan ini, sesuai dengan Petunjuk Dewan berikut ini serta sedapat mungkin sesuai dengan hukum negara-negara anggota Uni Eropa.

MK: Декларација за сообразност на ЕУ

Ние, Grundfos, изјавуваме под целосна одговорност дека произведете CU323, CU352, CU354, CU362, CU372, на кои се однесува допунаведената декларација, се во согласност со овие директиви на Советот за приближување на законите на земјите-членки на ЕУ.

NO: EUs samsvarsærklæring

Vi, Grundfos, erklærer under vårt eneansvar at produktene CU323, CU352, CU354, CU362, CU372 som denne erklæringen gjelder, er i samsvar med styrets direktiver om tilnærming av forordninger i EU-landene.

TH: คำประกาศความสอดคล้องตามมาตรฐาน EU

เราในฐานะของ บริษัท Grundfos ขอประกาศภายใต้ความรับผิดชอบของเราแต่เพียงผู้เดียวว่าผลิตภัณฑ์ CU323, CU352, CU354, CU362, CU372 ซึ่งเกี่ยวข้องกับคำประกาศนี้ มีความสอดคล้องกับระเบียบค่าส่งตามรายการด้านล่างนี้ของสภาราชชาชีว่าด้วยค่าประมาณค่าตามกฎหมายของรัฐที่เป็นสมาชิก EU

VI: Tuyên bố tuân thủ EU

Chúng tôi, Grundfos, tuyên bố trong phạm vi trách nhiệm duy nhất của mình rằng sản phẩm CU323, CU352, CU354, CU362, CU372 mà tuyên bố dưới đây có liên quan tuân thủ các Chỉ thị Hội đồng sau về việc áp dụng luật pháp của các nước thành viên EU.

UA: Декларація відповідності директивам ЕУ

Ми, компанія Grundfos, під нашу одноосібну відповідальність заявляємо, що вироби CU323, CU352, CU354, CU362, CU372, до яких відносяться нижченаведена декларація, відповідають директивам ЕУ, переліченим нижче, щодо тотожності законів країн-членів ЄС.

JP: EU 適合宣言

Grundfos は、その責任の下に、CU323, CU352, CU354, CU362, CU372 製品が EU 加盟諸国の法規に関連する、以下の評議会指令に適合していることを宣言します。

BS: Izjava o usklađenosti EU

Mi, kompanija Grundfos, izjavljujemo pod vlastitom odgovornošću da je proizvod CU323, CU352, CU354, CU362, CU372, na koji se odnosi izjava ispod, u skladu sa niže prikazanim direktivama Vijeća o usklađivanju zakona država članica EU.

KZ: Сәйкестік жөніндегі ЕО декларациясы

Біз, Grundfos, ЕО мүше елдерінің заңдарына жақын төменде көрсетілген Кеңес директиваларына сәйкес төмендегі декларацияға қатысты CU323, CU352, CU354, CU362, CU372 өнімдері біздің жеке жауапкершілігімізде екенін мәлімдейміз.

MY: Perisytiharan keakuran EU

Kami, Grundfos, mengisytiharkan di bawah tanggungjawab kami semata-mata bahawa produk CU323, CU352, CU354, CU362, CU372, yang berkaitan dengan perisytiharan di bawah, akur dengan Perintah Majlis yang disenaraikan di bawah ini tentang penghampiran undang-undang negara ahli EU.

AR: إقرار مطابقة EU

نحن نحن، جروندفوس، بمقتضى مسؤوليتنا الفردية بآن المتكبين، اللذين يخص بهما الإقرار أدناه، بكوننا مطابقيين لتوجيهات المجلس المذكورة أدناه بشأن التوافق (بين قوانين الدول أعضاء المجموعة الاتحاد الأوروبي).

TW: EU 合格聲明

基蘭富根據我們唯一的責任，茲聲明與以下聲明相關之 CU323, CU352, CU354, CU362, CU372 產品，符合下列近似 EU 會員國法律之議會指令。

AL: Deklara e konformitetit të BE

Ne, Grundfos, deklarojmë vetëm nën përgjegjësinë tonë se produktet CU323, CU352, CU354, CU362, CU372, me të cilat lidhet kjo deklarate, janë në pajtueshmëri me direktivat e Këshillit të renditura më poshtë për përfaqësimin e ligjeve të shteteve anëtare të BE-së.

- Low Voltage Directive (2014/35/EU).
Standard used:
EN 61010-1:2010
- EMC Directive (2014/30/EU).
Standard used:
EN61326-1:2013
- RoHS2 Directive (2011/65/EU and 2015/863/EU).
Standard used:
EN50581:2012

This EU declaration of conformity is only valid when published as part of the Grundfos safety instructions (publication number 97775216 1219).

Bjerringbro, 18/01/2017



Svend Aage Kaas
Director
Grundfos Holding A/S
Poul Due Jensens Vej 7
8850 Bjerringbro, Denmark

Person authorised to compile the technical file and empowered to sign the EU declaration of conformity.

中国 RoHS

产品中有害物质的名称及含量

部件名称	有害物质					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr6+)	多溴联苯 (PBB)	多溴联苯醚 (PBDE)
印刷电路板	X	O	O	O	O	O
紧固件	X	O	O	O	O	O
本表格依据 SJ/T 11364 的规定编制 O: 表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。 X: 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 该规定的限量要求。  该产品环保使用期限为 10 年，标识如左图所示。 此环保期限只适用于产品在安装与使用说明书中所规定的条件下工作						

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